



Scientific and Production Association
"KASKAD" Ltd.

Approved

Scientific and Production Association "KASKAD" Ltd.

Director General



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FIXED AEROSOL FIRE-EXTINGUISHING SYSTEM

(AF SYSTEM)

TECHNICAL DESCRIPTION AND

OPERATION INSTRUCTION

TD AFS KASKAD

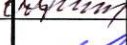


Saint-Petersburg
2016

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Chg	Sheet	Symbol	Sign	Date	Fixed aerosol fire-extinguishing system technical description and operation instruction	Lit.	Sheet	Sheets	
Designed	Zikov		28.03.2016	B			2	15	
Checked	Andryushenko		28.03.2016	SPA "KASKAD" Ltd.					
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Approved	Voronov		28.03.2016						

1. TECHNICAL DESCRIPTION

1.1 Introduction

The technical description and operation instruction is assigned to study the principle of operation and operating rules of a ship aerosol fire-extinguishing system (AFS) manufactured by Scientific and Production Association "KASKAD" Ltd.

The ship aerosol fire-extinguishing system ensures localization and fire extinction of volatile flammable and combustible liquids, solid inflammable materials and electrical equipments, including machinery under the voltage up to 40 kV, in closed spaces of different configuration and size.

Mode of the system functioning is based on the inhibition process of chemical reactions originating in the flame by fine-dispersed particles of fire-extinguishing aerosol. Aerosol is formed during operation of an aerosol fire-extinguishing generator and it fills in the protected space. When the generator stops discharging the agent, aerosol remains in suspension for 30-50 minutes, maintaining a concentration that excludes a possibility of re-ignition. Fire-extinguishing aerosol discharge does not cause a decrease of oxygen concentration in the premises.

1.2 AF system components

AF system consists of the following:

- Aerosol fire-extinguishing generators (SOT-1M, SOT-2M, SOT-2M-KV): the quantity and type of the generators is determined by calculations in accordance with the volume and description of the protected premises. AF generators, which are mounted directly in the protected premises, during operating discharge aerosol that is formed from solid aerosol-forming charge. Aerosol penetrating in the protected premises is evenly distributed along the volume and suppresses flaming;
- Control and signaling panel (CSP) is designed for the following purposes: to actuate the aerosol fire-extinguishing generator at “extinguishing” mode (actuation with automatic time delay) and at “Emergency actuation” mode (actuation without time delay); to give a signal to activate the warning alarm, to deactivate ventilation and other devices in the protected premises at the system actuation; to control integrity of the starting circuits with malfunction indication;
- Board of intermediate relays (BIR) (may not be applicable) is designed to control the warning alarm, to deactivate ventilation and other equipment (if available) in protected premises on command from the CSP;
- Audible and visual alarm (may not be applicable) is designed to give “AEROSOL, GO AWAY!”, a light and sound signal when the AF system is actuated (including explosion-proof);
- Connecting boxes (may not be applicable) and electric cables.

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2. OPERATION INSTRUCTION

AF system operating modes

The system has four operating modes:

- Standby mode;
- System monitoring mode;
- Actuation with time delay;
- Emergency actuation without time delay.

2.1 Standby mode

Standby mode is used at everyday system operation.

The AF system is to be in a complete operable state. At the same time:

- the starting circuits of generators are fault-free;
- the following circuits and devices in the protected premises are fault-free:
 - alarm of the AF system actuation;
 - ventilation and other equipment (if available) disconnection;
- supplied voltage meets the required voltage (24 ± 8 VDC) which may be supplied from all possible sources involved in AF system power supply, for instance:
 - shore-based power supply;
 - basic power supply;
 - battery supply;
 - battery supply with activated charger;
 - supply from emergency diesel-generator, etc;
- uninterrupted power supply from normal and emergency power sources is provided by means of ship's automatic devices;
- "POWER 24VDC CSP" and "POWER 24VDC BIR" indicators (if available) on the CSP front panel light, confirming that supply voltage is available in the input CSP and BIR terminals;
- automatic circuit breaker "POWER 24VDC" is enabled;
- the "POWER" tumbler is in the "OFF" position;
- the "START" tumblers on the CSP front panel are in the "OFF" position and sealed;
- AFS CSP cover is closed.

The mode is not time limited.

When giving over the watch, it's required to check power supply availability - indicators "POWER 24VDC CSP" and "POWER 24VDC BIR" must glow.

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2.2 System monitoring mode

System monitoring mode is used at AFS starting-up and adjustment works, acceptance tests and operational check.

AF system is at “standby mode”.

Open the CSP cover. Switch the “POWER” tumbler to the “ON” position. At the moment of power supply a built-in CSP buzzer is to sound. Within next several seconds the following message appears on LCD:

SPA «KASKAD»

AFS CSP

$N/G_1 - \dots - G_N$ $V \times \times$

where:

$N/G_1 - \dots - G_N$ - CSP configuration determining the quantity of **fire-extinguishing lines*** (N) and the quantity of generators in each line ($G_1 - \dots - G_N$);

$V \times \times$ - software version.

The CSP checks continuity of starting assemblies in automatic mode and alternately indicates all faulty circuits. At the absence of faults the following message appears on LCD lit with constant glow:

AF system
is operable

/

In the bottom left corner of LCD the rotating symbol «/» informs about normal operation of CSP.

If a faulty starting circuit is found, the built-in CSP buzzer starts to sound and the following message appear on LCDs:

L NN *<line name>*

High R start. circuit

G A

where:

NN - number of a faulty line;

G - number of a faulty circuit in the line NN;

A - symbol of automatic system monitoring mode.

* **Fire-extinguishing line (NN)** means one or several adjacent or integrated premises at which fire-extinguishing is carrying out simultaneously. When AF system is actuated all premises being a part of one line will be filled by fire-extinguishing aerosol.

Example

If the circuits №2 and №4 in line 01 break, the following message appears:

```
L01  <line name>
      High R start. circuit
02  04
                                           A
```

where:

- 01 - number of a faulty line;
- 02 04 - numbers of faulty circuits in line 01;
- A - symbol of automatic system monitoring mode.

When any of ◀, ▶, ▲, ▼ button is pressed, CSP switches to manual system monitoring mode, and symbol "A" changes to symbol "M" (manual mode). Subsequent pressing of ◀, ▶, ▲, ▼ allows to get access to the information about all faults. CSP can be turned to automatic system monitoring mode by pressing ↵.

To switch off/on the built-in buzzer press button  on the front panel of the CSP, at that "◀" symbol appears/ disappears on LCD.

In case of electrical circuits break accompanied by insulation current, CSP turns to safe mode to prevent AF generators from unauthorized actuation. After that the following message appears on LCD:

```
SAFE MODE IS ON
Leak current found
Control disabled
```

]To test the condition of the control circuits of warning alarm activation, and ventilation and other devices deactivation (if available by AFS project):

- activate ventilation and other devices that are deactivated in protected premises at AFS actuation;
- press down and hold   during the test.

At this moment the built-in CSP buzzer starts to sound and the following message appears on LCD:

```
ATTENTION! TEST
equipment switch off
and alarm switch on
TEST in XX sec
```

where:

XX - countdown time in seconds until control voltage is supplied to test the operation of the circuits and the equipment mentioned above.

On expiry of programmed time control voltage is supplied from CSP and the following message is displayed:

TEST

The result of the test is considered to be positive if a warning alarm of the system actuation turns on and ventilation and other devices (if available) are deactivated in the protected premises.

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2.3 Actuation system mode

Before AF system actuation IT IS NECESSARY to do the following:

- Close down hatches and portholes, close the doors and ventilation damper in the protected premises;
- If possible, cut off the current from electric equipments and switch off machinery in the protected premises;
- Leave the room.

2.3.1. AF system actuation test with time delay

Check light emission of the indicators “POWER 24VDC CSP” and “POWER 24VDC BIR” on the CSP front panel, which confirms that supply voltage is available in input terminals of CSP and BIR.

Open the CSP front cover. Switch the “POWER” tumbler to the “ON” position. In the presence of the interlock device of a chosen line, it is necessary to insert a key into the interlock device and to turn it clockwise to the “I” position.

Break the seal, move the backstop and turn the “START” tumbler of a chosen line to the “ON” position.

At this time the built-in buzzer starts to sound and the following message appears on LCD:

```
L NN  <line name>
      AF system start
      LEAVE THE SPACE
      XX sec
```

where:

NN - number of an actuated line;

XX - countdown time in seconds before starting impulses are supplied.

At the same time the warning alarm is to be activated and ventilation and other devices (if available) are to be deactivated in the protected premise(s) of the first line.

Cancel of AF system actuation is possible only during the time delay of actuation by switching the “START” tumbler or by turning the “POWER” to the “OFF” position.

At the moment of starting impulses are supplied, the built-in CSP buzzer switches to an even sound and the following message appears on LCD:

```
L NN  <line name>
      AEROSOL START!!!

      GA ... .. GD
```

where:

NN - number of an actuated line;

G_A G_D - numbers of generators actuated at the moment.

The generators in the protected premises actuate.

After generators actuation the following message appears on LCD:

```
L NN    <line name>
        AEROSOL is
        in the space
        XX   min
```

where:

NN - number of an actuated line;

XX - time delay in minutes (countdown from 50 to 0 minutes)

Countdown is kept up to 50 minutes when it is prohibited to activate ventilation or to do any other actions which may cause a decrease of aerosol concentration in the protected premises.

Afterwards the following message appears on LCD:

```
L NN    <line name>
        SWITCH OFF
        starting tumbler
```

where:

NN - number of an actuated line.

After the message appeared switch the AFS mode to “Standby mode” by switching the “START” and the “POWER” tumblers to the “OFF” position.

If a supplementary blocking unit for the activation of a chosen line is provided, turn the inserted key counterclockwise up to the stop at the “O” position and take out the key.

To actuate the AF system in several lines it is necessary to carry out the above listed actions for each chosen line.

2.3.2. AFS emergency actuation without time delay

AFS emergency actuation is needed when the CSP is out of order or when there is a need to actuate the system without time delay.

Check light emission of “POWER 24VDC CSP” and “POWER 24VDC BIR” indicators on the CSP front panel, which confirms that supply voltage is available in input terminals of CSP and BIR.

Open the CSP front cover. Switch the “POWER” tumbler to the “ON” position. In the presence of the interlock device of a chosen line, it is necessary to insert a key into the interlock device and to turn it clockwise to the “I” position.

Break the seal, move the backstop and turn the “START” tumbler of a chosen line to the “ON” position.

At the same time the warning alarm is to be activated and ventilation and other devices (if available) are to be deactivated in the protected premise(s) of chosen line.

For AFS emergency actuation push and hold for 1-2 sec each of suitable “EMERGENCY START” buttons one by one. As doing so, supply of starting impulses is carried out **without time delay**.

Pressing of the “EMERGENCY START” buttons does not influence the process of the CSP work. The CSP continues functioning in a current state counting the time delay until the moment of generators actuation. After the required time passes, the CSP supplies the impulses to all the starting circuits of the chosen line again.

ATTENTION

To avoid overloading of AFS power circuits it is strongly prohibited to:

- Simultaneously press the “EMERGENCY START” buttons for several times;
- Hold “EMERGENCY START” buttons when a following message appears on LCD:

```
L NN  <line name>
      AEROSOL START!!!

      GA ... .. GD
```

The following message appears on LCD after supply of starting impulses to all the circuits of the first line:

```
L NN          <line name>
      AEROSOL is
      in the space
      XX min
```

where:

NN - number of an actuated line;

XX - countdown time after actuation from 50 to 0 minutes.

Countdown is kept up to 50 minutes when it is prohibited to activate ventilation or to do any other actions which may cause a decrease of aerosol concentration in the protected premises filled with aerosol.

Afterwards a message appears on LCD:

L NN <line name>
SWITCH OFF
starting tumbler

where:

NN - number of an actuated line.

When the message appears, it is required to switch AFS to standby mode by turning the “START” and the “POWER” tumblers to the “OFF” position. If a supplementary blocking unit for the activation of a chosen line is provided, turn the inserted key counterclockwise up to the stop at the “O” position and take out the key

To actuate the AF system in several lines it is necessary to carry out the above listed actions for each chosen line.

3. SAFETY MEASURES AT AFS APPLICATION

Only people familiarized with the operation instructions as well as aerosol fire-extinguishing generators and the CSP operation instructions are allowed to apply AF system. Data on people names allowed to use AF system is written down in the record. (Annex 2).

Within 50 minutes after AFS actuation it is prohibited to activate ventilation or to do any other actions which may cause a decrease of aerosol concentration in the protected premises.

When aerosol discharge was completed, it is allowed to enter the protected premises but wearing respirators or any other respiratory apparatus protectors to avoid aerosol penetration into the respiratory tract.

The protected premises filled with aerosol have nil visibility.

In case of aerosol contact with eyes or mouth, it is necessary to wash them with fresh water without delay.

Aerosol can be cleaned away from the surface by water, dry or damp cloth, or by a vacuum cleaner.

When carrying out welding works or using any spark equipment (cutters, scarfing machines, etc.) in the protected premises, it is necessary to protect the AFS generators from sparks produced. If the works are carried out in close proximity with the generators, they should be demounted in order to prevent unauthorized actuation.

4. TECHNICAL MAINTENANCE

AFS technical maintenance implies compulsory operational check. If any malfunction is found, a corresponding note must be made in the AF system maintenance log (Annex 1). The malfunction must be immediately corrected (see i.5).

4.1 Monthly survey.

Monthly checking includes examination of AFS equipment mounting security, examination of starting assemblies integrity by switching the AF system to the “System monitoring” mode, and operation monitoring of the alarm, ventilation and other devices deactivation efficiency in the protected premises.

If malfunctions are found, they should be immediately corrected.

4.2 Annual survey.

Annual checking is carried out in the same way as monthly survey plus visual examination of electrical connection inside the control and signaling panel, board of intermediate relays, visual and audible alarm devices, connection boxes and the generators of AF system.

4.3 Regular survey.

Regular checking is implemented once in 5 (five) years in accordance with the “Program and method of regular survey of fixed aerosol fire-extinguishing system” or after replacement of the generators in scope of the “Program and method of acceptance test of aerosol fire-extinguishing system”.

AFS regular examination is to be conducted by the manufacturer or an organization certified by SPA “KASKAD” Ltd..

Results of all checks are to be registered in the AF system maintenance log (Annex 1).

ATTENTION!

Aerosol fire-extinguishing generators are disposable items and are not subjected to renewal or recharge.

Operating life of the generators is 5 years from the date when commissioning test being carried out at the ship. Total service life of the generators is 7,5 years from the manufactured date.

When the operating life expires (after 5 years), only subsequent to the results of the generators examination by the manufacturer (or an organization accredited by the manufacturer) it is possible to prolong the operating life of the generator till an expire date of its service life.

If you have any questions concerning generators delivery owing to their expiry date please do not hesitate to contact SPA “KASKAD” Ltd.

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5. POSSIBLE MALFUNCTIONS AND TROUBLESHOOTING METHODS

To correct AFS malfunctions is possible only when the CSP is disconnected from the circuit.

№	Malfunction type	Method of correction
1	In "System monitoring" mode LCD shows a number of faulty line and the built-in buzzer sound.	Switch off the power supply of CSP. Check the faulty line with a multimeter. Fix the faulty circuit. Switch the "POWER" tumbler to the "ON" position and make sure the circuit is nonfaulty.
2	In "System monitoring" mode LCD shows the message that CSP is in safe mode.	Current leakage is found out. Check all external circuits of CSP on the subject of possible contact with the case. Check insulation resistance of all external circuits and aerosol fire-extinguishing generators.
3	The warning alarm does not switch-on and ventilation and other equipment does not switch off while holding  .	There are self-repairing protection devices in the control circuits of the CSP. Their operation can be caused either by short circuit in the external circuit or by connecting other electric devices which consumption current is more than 1A. The protecting devices self-repair at completely dead circuits within 30-60 seconds. To correct: - Disconnect the faulty line from the CSP. Having waited for 30-60 seconds, check availability of voltage in the output terminal of the CSP after holding  buttons pressed. - Check the disconnected faulty control circuit for circuit discontinuity and short circuit. If the circuit is operative, then connect it to the CSP. - Check whether BIR and control devices are operative.
4	"PROTECTION" sign appears when the "POWER" tumbler is on.	Power voltage supplied is too high. Switch the "POWER" tumbler to the "OFF" position. Supply the required power voltage.
5	Automatic circuit breaker "POWER 24 VDC" switches off	Overcurrent protection is activated. The possible reasons for this are: - too high power voltage (safety circuit is enabled); - wrong actions during AFS emergency start (see paragraph 5.3.2).

If any malfunction is found out, a person in charge is to write it down in the maintenance log (Annex 1).

6. AF SYSTEM MAINTENANCE LOG

[illegible]

7. AF SYSTEM INSTRUCTION LOG

Name/Surname and position of a person trained to use AFS, date.	Signature	Trainer's signature